

Instituto Tecnológico de Costa Rica

Campus Tecnológico Local San José

Laboratorio 09: Seguridad

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Fecha de entrega:

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1. GE es el propietario de la tabla employee y department.

a. Otorgue a CU permisos para insertar en la tabla employee. 5pts.

Primero se crea el esquema CU:

```
-- Crear el esquema CU
CREATE USER CU
  IDENTIFIED BY cu
  DEFAULT TABLESPACE GE_Data
  QUOTA 10M ON GE_Data
  TEMPORARY TABLESPACE temp
  QUOTA 5M ON SYSTEM
  QUOTA 10M ON GE_Ind;

GRANT CONNECT to CU;
```

User CU created.

Grant succeeded.

Después se le da permiso a CU para insertar en la tabla employee:

```
GRANT INSERT ON employee TO CU;
```

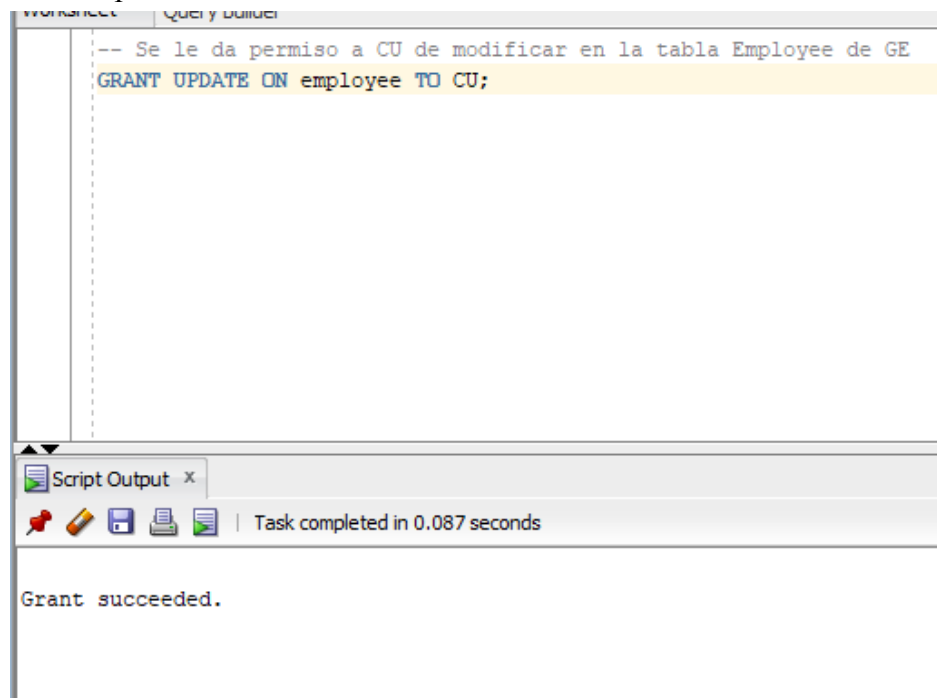
Script Output x

Task completed in 0.131 seconds

Grant succeeded.

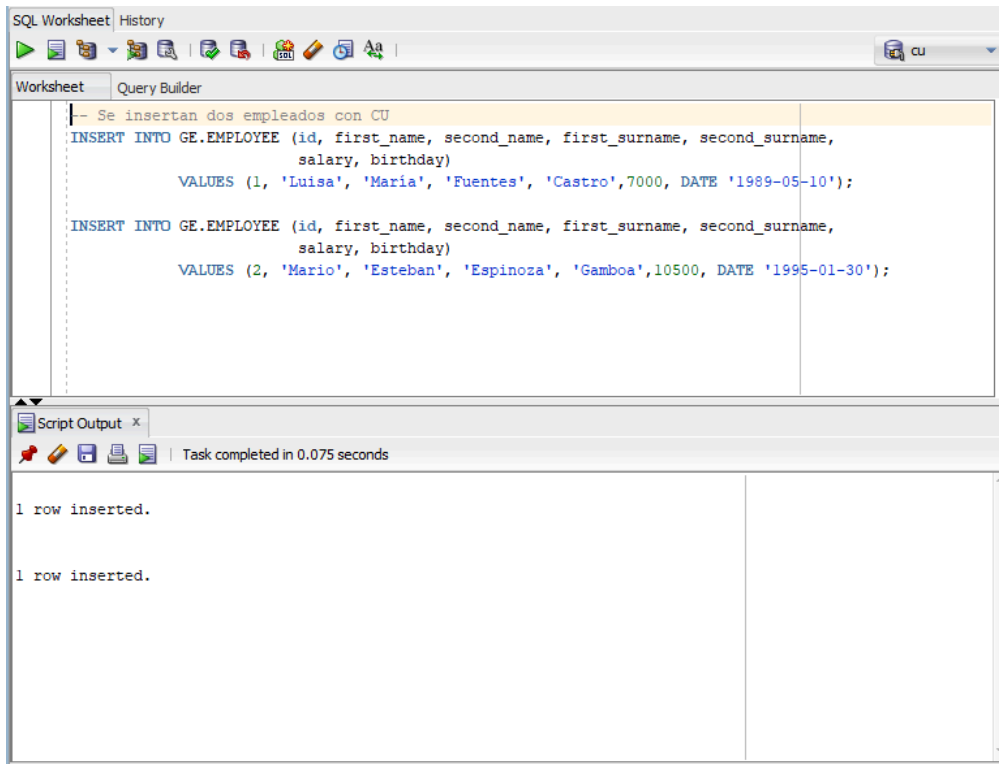
b. Otorgue a CU permisos para modificar en la tabla employee. 5pts.

Se le da permiso a CU de modificar la tabla:



c. Inserte 2 empleados en la tabla employee con CU. 5pts.

Se agregan dos empleados a la tabla con el usuario CU. Se puede ver en la parte de arriba a la derecha que se utilizó este usuario:

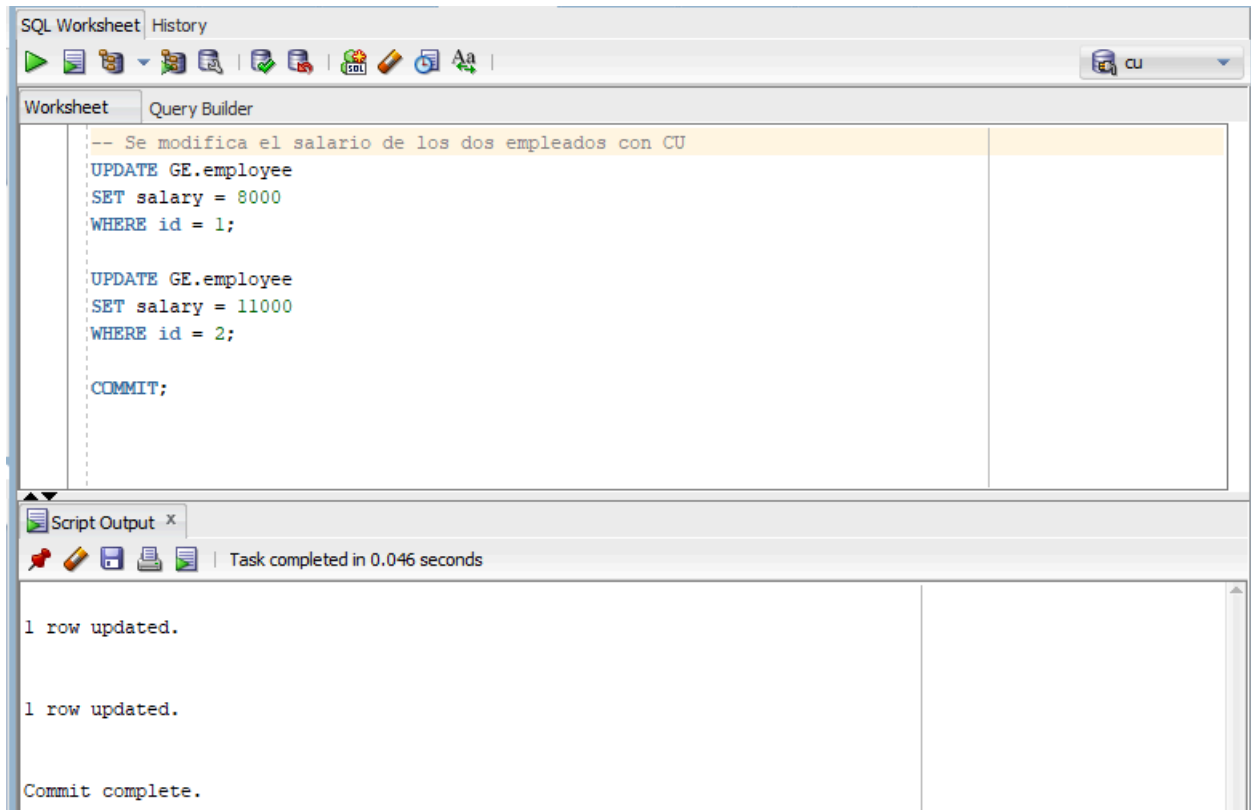


Aquí se ven los datos en la tabla:

ID	FIRST_NAME	SECOND_NAME	FIRST_SURNAME	SECOND_SURNAME	SALARY	BIRTHDAY
1	1 Luisa	Maria	Fuentes	Castro	7000	10-MAY-89
2	2 Mario	Esteban	Espinoza	Gamboa	10500	30-JAN-95

d. Modifique el salario de estos empleados con CU. 5pts.

Se modifica el salario de los empleados con CU. Se puede ver en la parte de arriba a la derecha que se utilizó este usuario:

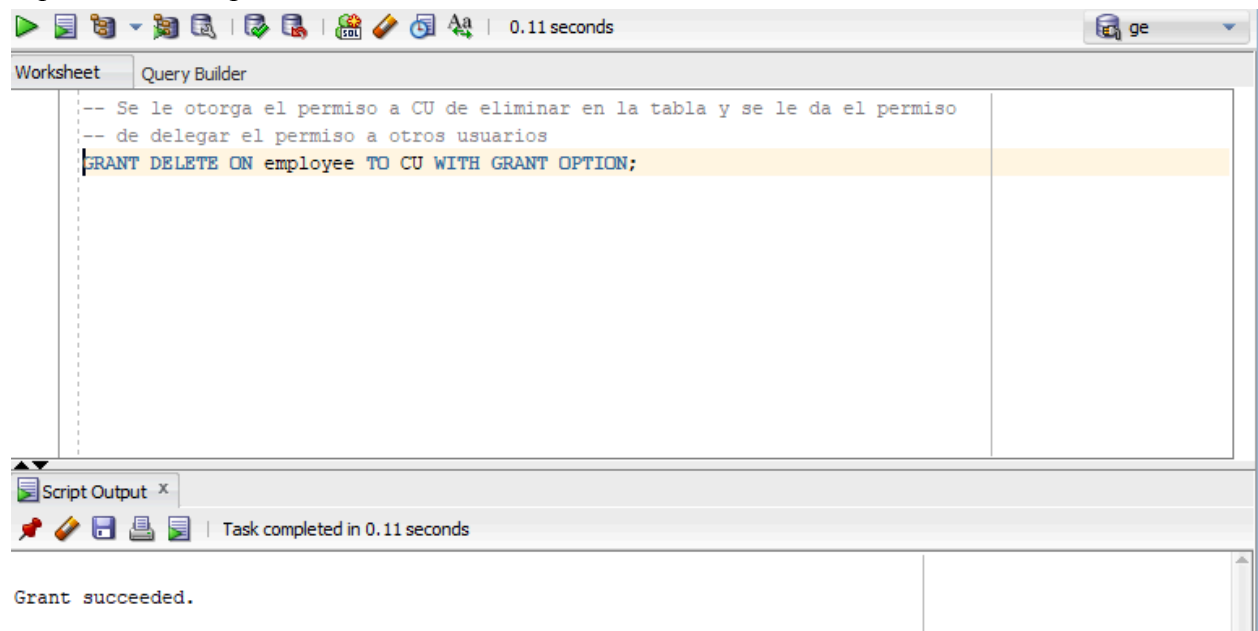


Así se ve la tabla ahora:

ID	FIRST_NAME	SECOND_NAME	FIRST_SURNAME	SECOND_SURNAME	SALARY	BIRTHDAY
1	Luisa	María	Fuentes	Castro	8000	10-MAY-89
2	Mario	Esteban	Espinoza	Gamboa	11000	30-JAN-95

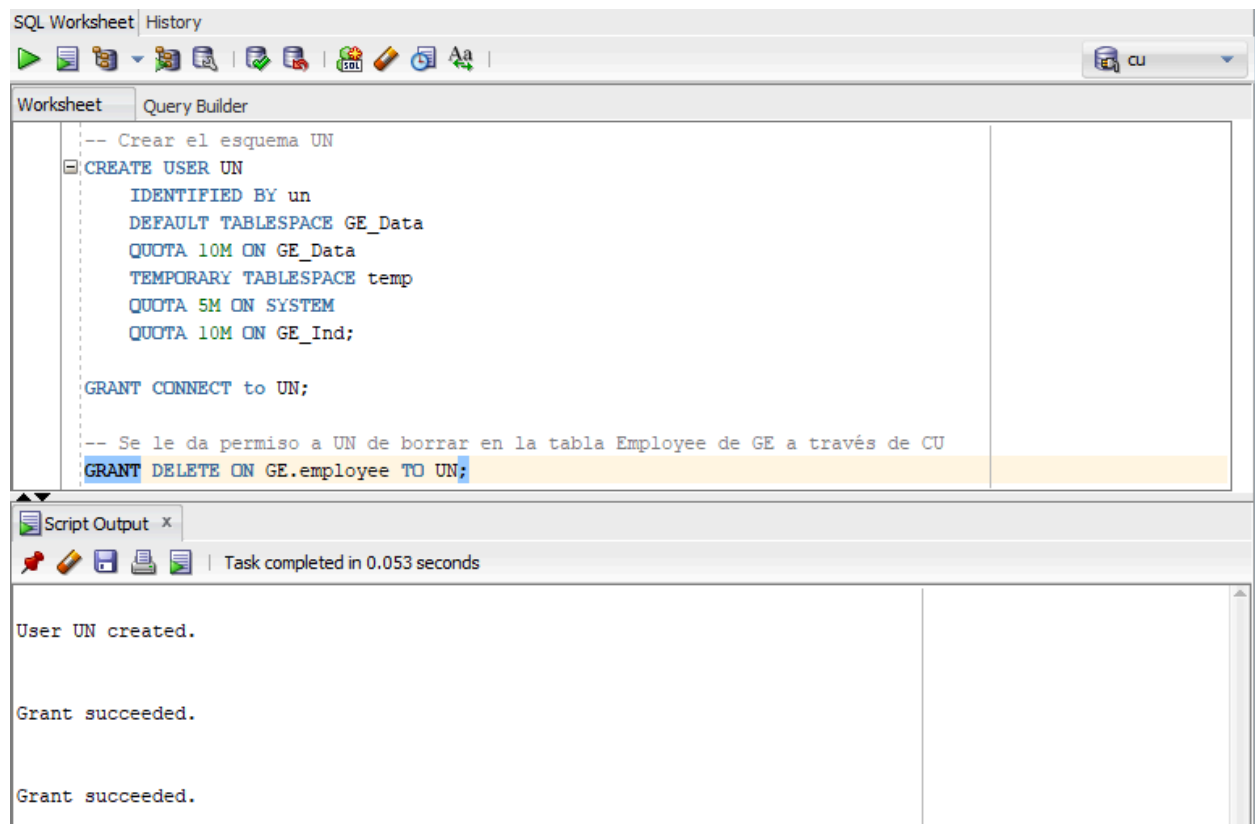
e. Otorgue a CU permisos para eliminar en la tabla employee con la opción de delegar el permiso a otros usuarios. 10pts.

Aquí GE le da el permiso a CU:



f. Otorgue a UN permisos para eliminar en la tabla employee con CU y elimine un empleado. 10pts.

Primero se crea UN y después CU (se puede ver en la parte de arriba a la derecha) le permite a UN borrar un empleado:



The screenshot shows an SQL Worksheet window with a toolbar at the top. The 'Worksheet' tab is active, and the 'Query Builder' pane is empty. The script in the main editor is as follows:

```
-- Crear el esquema UN
CREATE USER UN
  IDENTIFIED BY un
  DEFAULT TABLESPACE GE_Data
  QUOTA 10M ON GE_Data
  TEMPORARY TABLESPACE temp
  QUOTA 5M ON SYSTEM
  QUOTA 10M ON GE_Ind;

GRANT CONNECT TO UN;

-- Se le da permiso a UN de borrar en la tabla Employee de GE a través de CU
GRANT DELETE ON GE.employee TO UN;
```

Below the script editor is a 'Script Output' window. It shows the results of the executed script:

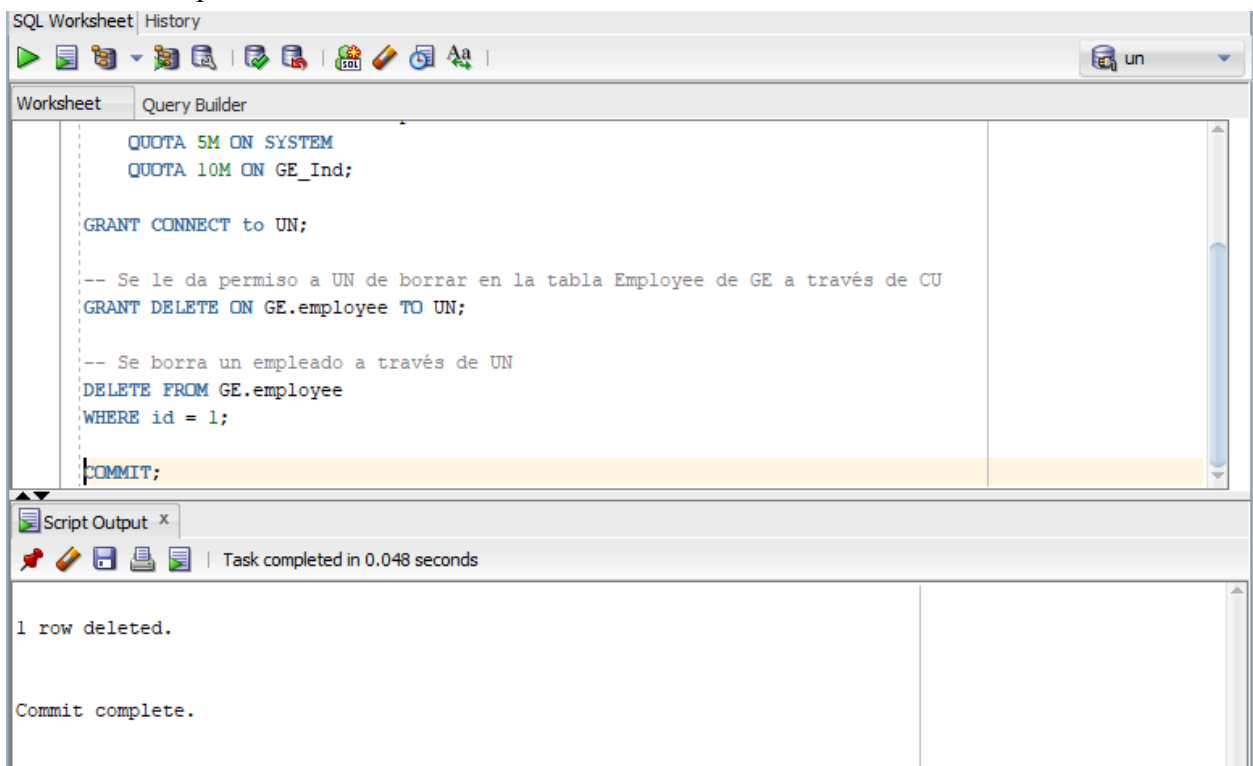
```
User UN created.

Grant succeeded.

Grant succeeded.
```

The 'Task completed in 0.053 seconds' message is visible above the output window.

Se borra un empleado a través de UN:



The screenshot shows an SQL Worksheet interface. The top toolbar includes icons for running queries, saving, and undo. The 'Worksheet' tab is active, showing a 'Query Builder' view. The SQL script in the editor is as follows:

```
QUOTA 5M ON SYSTEM
QUOTA 10M ON GE_Ind;

GRANT CONNECT to UN;

-- Se le da permiso a UN de borrar en la tabla Employee de GE a través de CU
GRANT DELETE ON GE.employee TO UN;

-- Se borra un empleado a través de UN
DELETE FROM GE.employee
WHERE id = 1;

COMMIT;
```

Below the editor, the 'Script Output' tab is active, displaying the results of the query execution:

```
1 row deleted.

Commit complete.
```

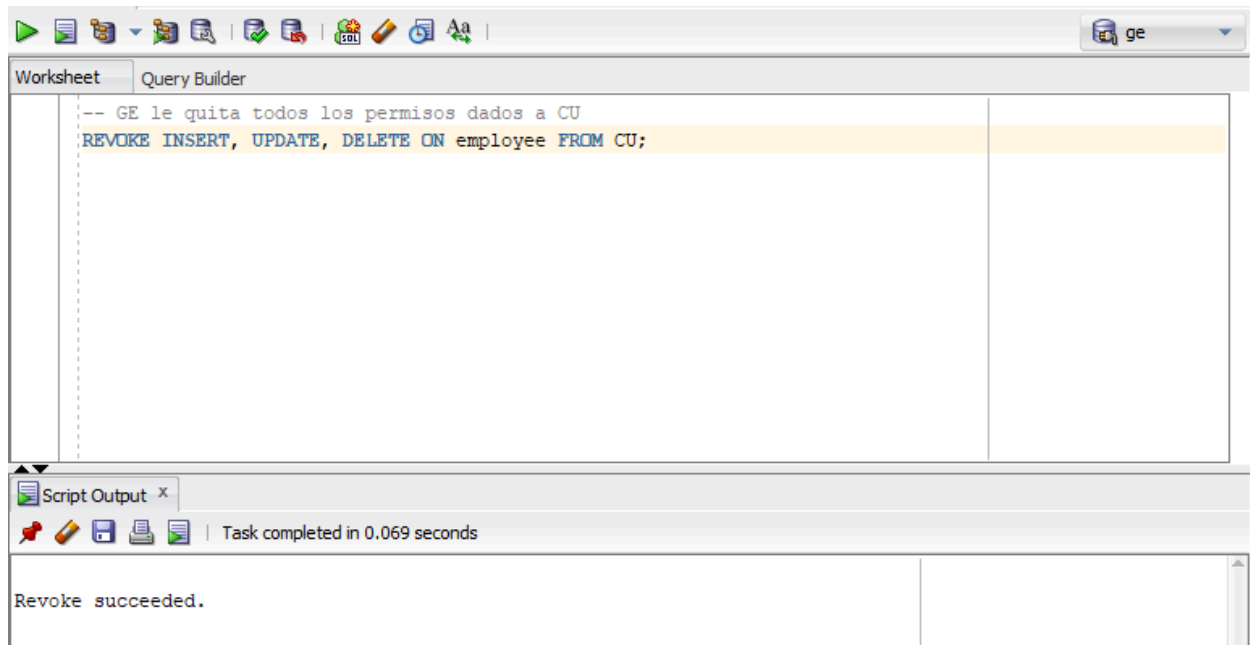
The output also indicates that the task was completed in 0.048 seconds.

Así se ve ahora la tabla:

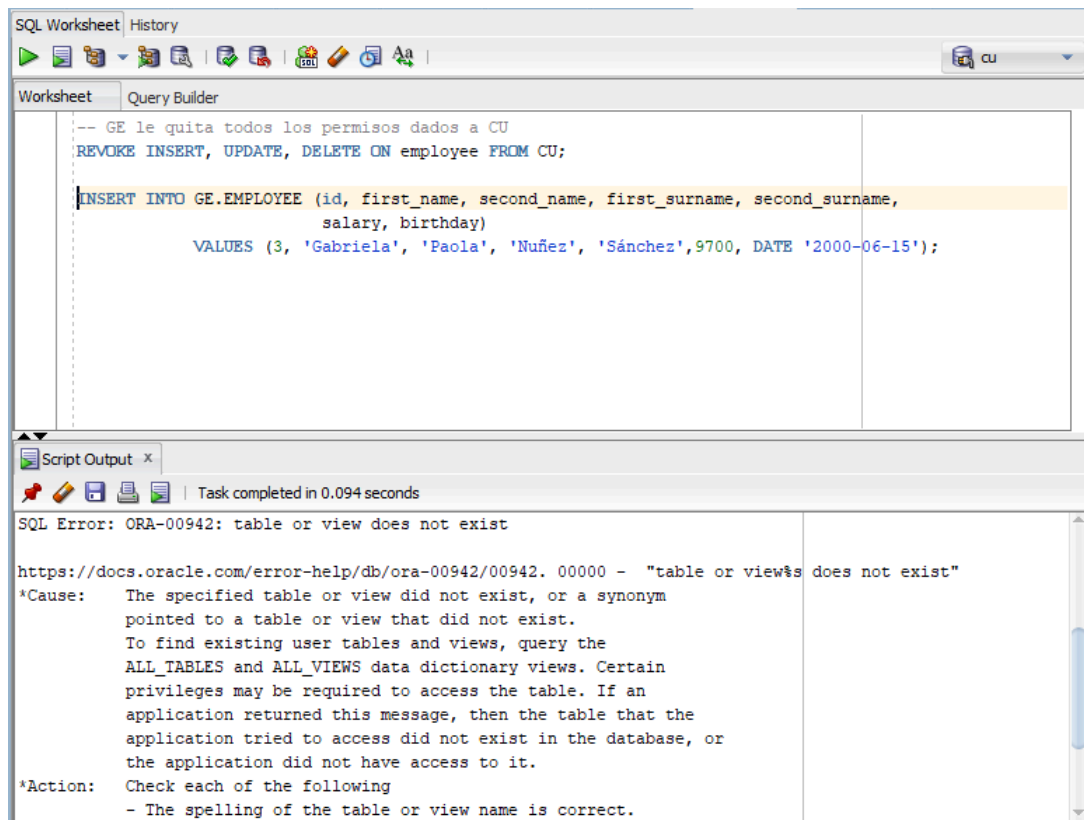
ID	FIRST_NAME	SECOND_NAME	FIRST_SURNAME	SECOND_SURNAME	SALARY	BIRTHDAY
1	2 Mario	Esteban	Espinoza	Gamboa	11000	30-JAN-95

g. Deniegue los permisos otorgados a CU. 10pts.

GE le quita todos los permisos a CU que le dió:



Ahora CU no puede ingresar un empleado:



Si intento borrar un empleado desde UN, tampoco puedo:

The screenshot shows an SQL Worksheet interface with a 'Query Builder' tab. The script contains the following SQL commands:

```
-- GE le quita todos los permisos dados a CU
REVOKE INSERT, UPDATE, DELETE ON employee FROM CU;

-- Se intenta insertar un empleado con CU
INSERT INTO GE.EMPLOYEE (id, first_name, second_name, first_surname, second_surname,
                        salary, birthday)
VALUES (3, 'Gabriela', 'Paola', 'Nuñez', 'Sánchez', 9700, DATE '2000-06-15');

-- Se intenta borrar un empleado con UN
DELETE FROM GE.employee
WHERE id = 2;
```

The 'Script Output' tab shows the execution results, including an error message:

```
Error starting at line : 10 in command -
DELETE FROM GE.employee
WHERE id = 2
Error at Command Line : 10 Column : 16
Error report -
SQL Error: ORA-00942: table or view does not exist

https://docs.oracle.com/error-help/db/ora-00942/00942. 00000 - "table or view does not exist"
Cause: The specified table or view did not exist, or a synonym
pointed to a table or view that did not exist.
To find existing user tables and views, query the
ALL_TABLES and ALL_VIEWS data dictionary views. Certain
```